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Project Expo 2026

Forensic Sketch Recognition System

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Abstract

This project presents an AI-based Forensic Sketch Recognition System to help law enforcement identify suspects from eyewitness sketches when photographic evidence is unavailable. Traditional face recognition systems perform poorly on sketches due to lack of texture and detail. To overcome this, the system uses a hybrid approach combining appearance-based, geometric, and pose-invariant feature extraction techniques. Histogram of Oriented Gradients (HOG) extracts appearance features, while facial landmark detection captures geometric relationships, and pose-invariant facial ratios handle orientation variations. The system includes a user-friendly sketch generation module and a recognition module that matches sketches with a criminal database, retrieving and ranking top suspects. It achieves around 85% accuracy with an average processing time of 1.9 seconds, making it reliable and efficient.

1. Introduction

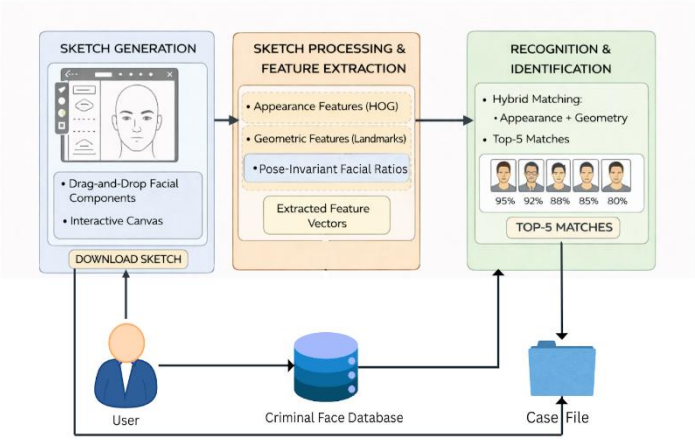
In criminal investigations, suspect identification often relies on eyewitness sketches when photographic evidence is unavailable. Manually matching these sketches with large databases is time-consuming and prone to errors, while traditional face recognition systems perform poorly due to the lack of detailed features in sketches.
To address this, the project develops an AI-based Forensic Sketch Recognition System that improves identification accuracy and speed. It uses a hybrid approach combining HOG-based appearance features, facial landmark-based geometric features, and pose-invariant techniques to match sketches with a database and retrieve the most likely suspects.

2. Methodology / Design Approach

The proposed system follows a structured pipeline for sketch-based suspect identification. It uses React and Fabric.js for sketch generation and Python with OpenCV, dlib, and scikit-learn for recognition.The process starts with sketch creation using a drag-and-drop interface, followed by preprocessing to normalize and clean the image.

Features are extracted using HOG for appearance and facial landmarks for geometric details, along with pose-invariant ratios to handle orientation changes.

These features are then compared with a database using a hybrid matching algorithm. The system calculates similarity scores and retrieves the top matching suspects, providing a ranked Top-5 result.



3. Results and Discussion

The system successfully performs sketch-to-photo matching using a hybrid feature extraction approach. Tested on around 37,000 images, it achieved about 85% accuracy with an average processing time of 1.9 seconds, effectively retrieving and ranking top matching suspects. Results show that combining HOG, facial landmarks, and pose-invariant techniques improves accuracy compared to single methods. The system works best for clear sketches, though performance may slightly drop for low-quality or extreme pose variations. Overall, it provides a reliable and efficient solution for automated suspect identification.

4. Conclusion

The proposed system effectively identifies suspects using sketch-to-photo matching by combining appearance and geometric features, improving accuracy over traditional methods. With around 85% accuracy and 1.9 seconds processing time, it proves to be an efficient and practical tool for real-world forensic applications.

References

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